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To cite this article: Carl Emil Mørch Nielsen *et al* 2025 *2D Mater.* **12** 045015

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RECEIVED  
18 June 2025REVISED  
11 August 2025ACCEPTED FOR PUBLICATION  
8 September 2025PUBLISHED  
30 September 2025

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# From dark to bright: first-principles prediction of ground state excitons and trions in Janus transition metal dichalcogenide bilayers

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E-mail: [gabriel.bester@uni-hamburg.de](mailto:gabriel.bester@uni-hamburg.de)**Keywords:** excitons, optoelectronics, trions, transition metal dichalcogenide, Janus, bright ground stateSupplementary material for this article is available [online](#)

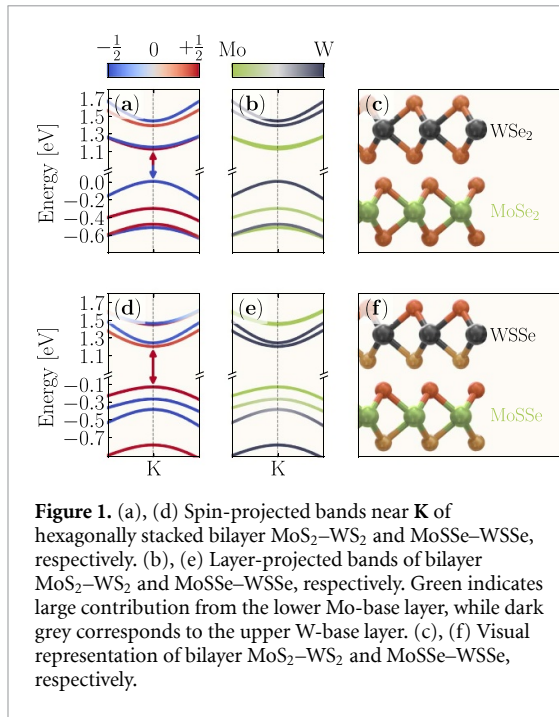
## Abstract

We investigate bilayer Janus transition metal dichalcogenides using *ab initio* many-body screened configuration interaction calculations and find that MoS<sub>2</sub>–WSe<sub>2</sub> and WSe<sub>2</sub>–WSe<sub>2</sub>, with Se–S interfaces, exhibit spin-allowed interlayer exciton and trion ground states, in contrast to the spin-forbidden ground states of conventional transition metal dichalcogenides bilayers. This behavior arises from intrinsic structural asymmetry and interface-induced polarization, which rearrange the band structure. We further show that, besides the intrinsic asymmetry, also the applied external strain enables tuning of the ground state exciton brightness, allowing for detailed control of their optical properties.

## 1. Introduction

The two-dimensional (2D) family of transition metal dichalcogenides (TMDs) is an interesting and widely-known platform for studying optoelectronics and excited states [1–3]. They feature significant bandgaps, large spin–orbit coupling effects, and intriguing characteristics when stacked vertically via van der Waals interaction [4–6]. In particular, bilayer heterostructures usually lead to type-II band alignments, where band edge electrons and holes localize in different monolayers [7, 8]. This leads to long-lived interlayer excitons with lifetimes in the nano-second range [9–11]. A popular material combination hosting these type of excitons is MoX<sub>2</sub>–WX<sub>2</sub> (X = S, Se, Te), where the electron is localized in MoX<sub>2</sub> and the hole in WX<sub>2</sub>. This can be seen already at the single-particle level in figure 1(b), where we color coded the band structure according to the localization of the state; with dark grey for WSe<sub>2</sub> and green for MoSe<sub>2</sub>. These systems are particularly notable for their lack of mirror and inversion symmetry and a negligible lattice mismatch between the layers,

typically around 0.1%, which prevents the formation of complex moiré patterns [12]. However, regardless of the stacking configuration, the exciton ground state remains spin-forbidden and therefore dark [11, 13, 14]—mainly an artifact of the marginal conduction band splitting in MoX<sub>2</sub>. To circumvent this issue, we propose bilayer Janus TMDs, which, as opposed to their conventional TMD counterparts, feature different chalcogen elements on either face, as shown in figure 1(f) [15, 16]. This heterogeneity instigates potential drops in the individual monolayers and, for some stacking configurations, at the interface between them—substantially altering the electronic properties of the materials [17–21]. The introduction of the different interfaces, combined with various material combinations and stacking orders, means that a wide range of designs are feasible. In total, we examined 181 unique combinations, and identified at least two promising Janus heterostructures, namely bilayer MoS<sub>2</sub>–WSe<sub>2</sub> and WSe<sub>2</sub>–WSe<sub>2</sub> with a Se–S interface, which exhibit a spin-allowed lowest energy transition along with a significant conduction band splitting. Figures 1(d) and (e) reveal the band



**Figure 1.** (a), (d) Spin-projected bands near  $\mathbf{K}$  of hexagonally stacked bilayer MoS<sub>2</sub>-WSe<sub>2</sub> and MoSSe-WSeSe, respectively. (b), (e) Layer-projected bands of bilayer MoS<sub>2</sub>-WSe<sub>2</sub> and MoSSe-WSeSe, respectively. Green indicates large contribution from the lower Mo-base layer, while dark grey corresponds to the upper W-base layer. (c), (f) Visual representation of bilayer MoS<sub>2</sub>-WSe<sub>2</sub> and MoSSe-WSeSe, respectively.

ordering near  $\mathbf{K}$  for MoSSe-WSeSe. Due to the potential drops instigated by the interface, the bands of the monolayers shift relative to each other (the dark grey WSeSe bands shift down and the green MoSSe bands shift up in energy), such that the upper most valence bands have MoSSe character, while the lower most conduction bands are of WSeSe character—opposite to the case of MoSe<sub>2</sub>-WSe<sub>2</sub> shown in figure 1(b). This reordering of the valence bands leads to a change of spin character of the uppermost valence band, from spin down in MoSe<sub>2</sub>-WSe<sub>2</sub> (blue in figure 1(a)) to spin up in MoSSe-WSeSe (red in figure 1(d)). The new ordering of the conduction bands does not change the spin character of the lowest conduction band state at  $\mathbf{K}$  (spin-up, red, in both figures 1(a) and (b)) but increases the conduction band splitting. This leads to a spin-allowed and potentially bright transition in MoSSe-WSeSe with a rather large conduction band splitting.

In this work, we show that bilayer Janus TMD heterostructures, specifically MoSSe-WSeSe and WSeSe-WSeSe with Se-S interfaces, exhibit spin-allowed ground state interlayer excitons and trions, in marked contrast to the spin-forbidden and optically dark ground states of conventional TMD bilayers. We attribute this behavior to the combined effects of intrinsic structural asymmetry and interface-induced polarization, which reorder the single-particle bands and enable optically active ground state transitions with substantial dark-bright energy splitting. Furthermore, we show that both intrinsic asymmetry and the application of external strain provide effective means to tune the brightness of the ground state excitons and trions, enabling detailed

control over their optical properties and excitonic fine structure.

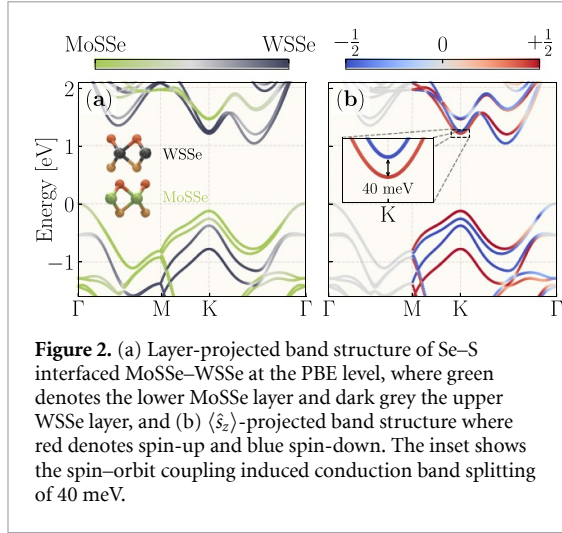
## 2. Results

We have conducted extensive calculations for both Se-S-interfaced bilayer MoSSe-WSeSe and WSeSe-WSeSe, both exhibiting unique yet similar properties. For the sake of brevity, the results discussed in this section focus exclusively on the bilayer MoSSe-WSeSe system, while findings for bilayer WSeSe-WSeSe are provided in the supplemental material (SM).

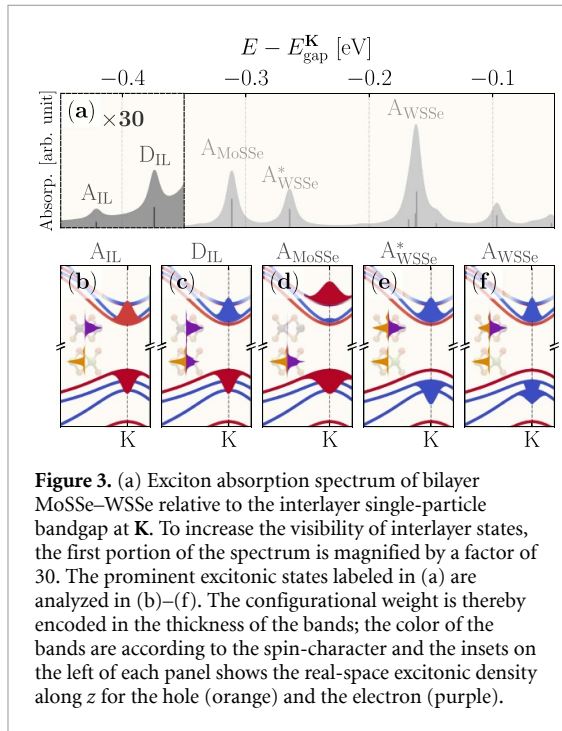
Figures 2(a) and (b) shows the band structure of Se-S interfaced MoSSe-WSeSe projected onto the constituting layers in (a) and with the spin-projection in (b), in a similar fashion to figures 1(d) and (e) but for the entire Brillouin zone. Notably, the material features type-II band alignment, where the lower most conduction bands at  $\mathbf{K}$  stem from the upper WSeSe layer, while the upper most valence bands at  $\mathbf{K}$  arise from the lower MoSSe layer [18, 22]. We find a lattice constant,  $a_0$ , of 3.22 Å [17, 23] and an interlayer distance,  $d$ , (metal-metal  $z$ -distance) of 6.29 Å. The spin-orbit coupling induced splittings at  $\mathbf{K}$  of the conduction and valence bands are 40 and 136 meV, respectively, and the energetically lowest direct transition is spin-allowed and lies exactly at  $\mathbf{K}$ . Moreover, the second and third highest valence bands are strongly hybridized between the two layers, as can be seen by the intermediate color of the bands in figure 2(a). We note, that the conduction band minimum lies around halfway between  $\mathbf{K}$  and  $\Gamma$ , sometimes denoted  $\mathbf{Q}$  [24–26], whereas the valence band maximum is situated at  $\Gamma$ . However, our focus lies on the low temperature limit, i.e. excitons with vanishing center-of-mass momentum,  $\mathbf{Q}_{\text{COM}} = 0$ , meaning only momentum allowed transitions around  $\mathbf{K}$  are relevant.

The exciton absorption spectrum of bilayer MoSSe-WSeSe is displayed in figure 3(a) relative to the single-particle direct bandgap at  $\mathbf{K}$ . The energetically lowest state, denoted  $A_{\text{IL}}$ , is a pure interlayer state, where the electron resides in WSeSe and the hole in MoSSe, and has an exciton binding energy of 0.31 eV (with respect to the single-particle bandgap at  $\mathbf{K}$ ). It should be noted that figure 3(a) shows results for a  $k$ -point sampling of the Brillouin zone using a Monkhorst-Pack grid of  $45 \times 45$ , which is not fully converged. However, extrapolating the results to the converged grid is straight forward in this case (see SM) and the binding energies reported throughout the text are the converged values.

As seen in figure 3(b), the constituents of  $A_{\text{IL}}$  are spin-aligned (electrons and holes both in red color), and the optical transition relatively bright. Interestingly,  $D_{\text{IL}}$  with a binding energy of 0.27 eV appears comparably brighter in the spectrum, even though the constituents seem spin-forbidden, as



**Figure 2.** (a) Layer-projected band structure of Se-S interfaced MoSSe-WSSe at the PBE level, where green denotes the lower MoSSe layer and dark grey the upper WSSe layer, and (b)  $\langle \hat{s}_z \rangle$ -projected band structure where red denotes spin-up and blue spin-down. The inset shows the spin-orbit coupling induced conduction band splitting of 40 meV.



**Figure 3.** (a) Exciton absorption spectrum of bilayer MoSSe-WSSe relative to the interlayer single-particle bandgap at  $\mathbf{K}$ . To increase the visibility of interlayer states, the first portion of the spectrum is magnified by a factor of 30. The prominent excitonic states labeled in (a) are analyzed in (b)–(f). The configurational weight is thereby encoded in the thickness of the bands; the color of the bands are according to the spin-character and the insets on the left of each panel shows the real-space excitonic density along  $z$  for the hole (orange) and the electron (purple).

portrayed in figure 3(c) (electron blue, hole red). This can be attributed to the electron localizing in both layers (purple electron density distributed over both layers, see inset of figure 3(c)), which allows for some overlap between electron and hole within the MoSSe layer. In essence, while  $A_{IL}$  is spin allowed but spatially indirect,  $D_{IL}$  is weakly spatially direct, and primarily spin-forbidden, although a weak spin-allowed admixing occurs. Furthermore, the bands are not fully spin-polarized, i.e.  $\langle \hat{s}_z \rangle \neq \pm 1/2$ , making the transition less spin-forbidden than anticipated by the different colors of the two bands in figure 3(c). Further details on the intralayer character of  $D_{IL}$  can be found in the SM. Due to the interlayer character of  $A_{IL}$  and the spin-asymmetry of  $D_{IL}$ , only minor electron-hole exchange coupling occurs for

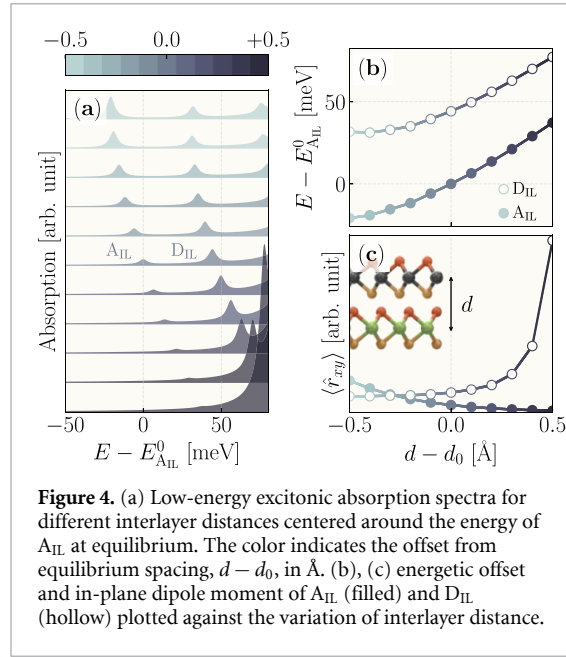
both states [27, 28]. This statement can be summarized as

$$\langle i\alpha | v | \alpha i \rangle = \begin{cases} \langle l_1 l_2 | v | l_2 l_1 \rangle \approx 0, & \text{for } A_{IL} \\ \langle \downarrow \uparrow | v | \uparrow \downarrow \rangle \approx 0, & \text{for } D_{IL} \end{cases}, \quad (1)$$

where  $i$  and  $\alpha$  denote electron (conduction) and hole (valence) states, respectively. Similarly,  $l_{12}$  indicate the layer of each constituent fermion, while the arrows symbolize the spin character. For  $A_{IL}$ , the spatial separation leads to a minute orbital overlap between the hole and electron wavefunctions, and for  $D_{IL}$ , the orthogonality of the spin wavefunctions leads to vanishing coupling. As such, the splitting between the two states (48 meV) is comparable to the conduction band splitting (40 meV). Pang and Wang [29] report comparable interlayer exciton binding energies of 0.32 eV for rhombohedral-stacked Se-S interfaced MoSSe-WSSe; however, they observe that the lowest-lying exciton is spin-forbidden and dark. This can be understood when realizing that the rhombohedral-stacked structure can be obtained by rotating one of our layers by  $180^\circ$  in figure 1(f). Such a rotation leads to an interchange of  $\mathbf{K}$  and  $\mathbf{K}'$ , and hence an inversion of the spin character in the bands associated with the rotated layer.

Figures 3(d)–(f) shows the three spin-allowed bright states that are most prominent in the intralayer part of the spectrum.  $A_{MoSSe}$  is a pure intralayer state residing in the MoSSe layer, and, when compared to the MoSSe single-particle bandgap at  $\mathbf{K}$  (found between the upper most valence band and the third conduction band), features an exciton binding energy of 0.45 eV. In contrast,  $A_{WSSe}^*$  and  $A_{WSSe}$  show intralayer WSSe character with exciton binding energies of 0.41 and 0.30 eV, respectively. Due to the mixed layer localization of the second and third highest valence bands we noted earlier in figure 2(a), the hole hybridizes between the two monolayers for both states (seen in the orange hole distribution). However,  $A_{WSSe}$  has a larger contribution from the third highest valence band compared to  $A_{WSSe}^*$ , which explains the smaller binding energy.

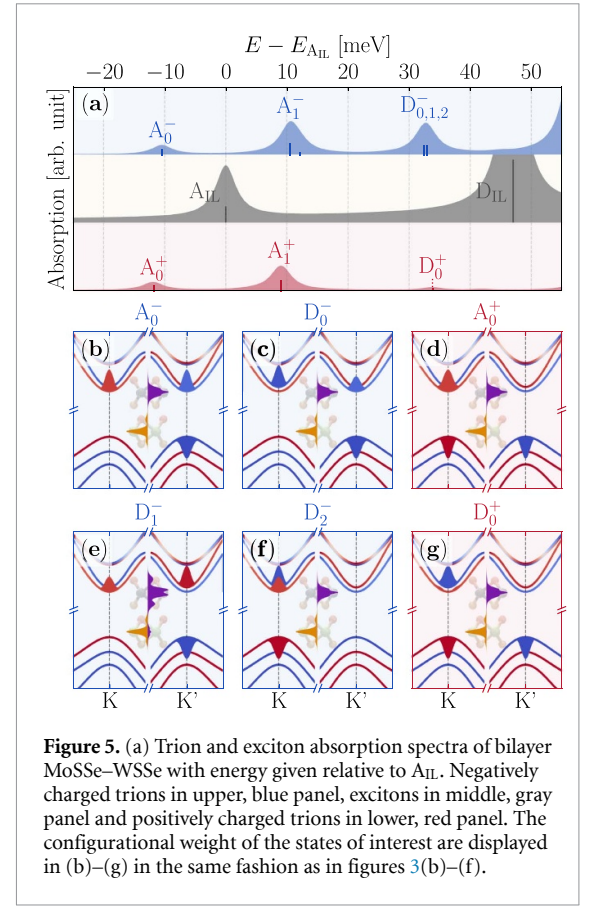
The presented excitonic states and associated binding energies compare well to their conventional TMD counterparts, namely  $MoS_2$ - $WS_2$  and  $MoSe_2$ - $WSe_2$ . Here, the interlayer binding energies are typically in the range of 0.3–0.4 eV, while intralayer states are bound by 0.3–0.6 eV [11, 13, 14, 30]. However, the splitting between the interlayer,  $D_{IL}$ , and intralayer states,  $A_{MoSSe}$ , (60 meV) is significantly smaller than for the regular TMD counterpart, e.g.  $MoS_2$ - $WS_2$  (200 meV) [11, 14, 30]. This can be attributed to the relatively small energetic offsets between the MoSSe and WSSe bands in the upper most valence bands compared to  $MoX_2$ - $WX_2$  (see figures 1(b) and (e))—an effect of the dipole moment induced by the Janus structured interface.



**Figure 4.** (a) Low-energy excitonic absorption spectra for different interlayer distances centered around the energy of  $A_{IL}$  at equilibrium. The color indicates the offset from equilibrium spacing,  $d - d_0$ , in Å. (b), (c) energetic offset and in-plane dipole moment of  $A_{IL}$  (filled) and  $D_{IL}$  (hollow) plotted against the variation of interlayer distance.

We have further investigated the effects of out-of-plane strains on the lowest-lying excitonic states,  $A_{IL}$  and  $D_{IL}$ , and depicted the results in figure 4(a). While previous work has explored the impact of in-plane strain and hydrostatic pressure [29, 31], our results specifically examine out-of-plane strain effects. Here, we vary the interlayer spacing (metal–metal distance),  $d$ , between the two layers around the equilibrium distance,  $d_0$ , and observe in figure 4(b) that both  $A_{IL}$  and  $D_{IL}$  experience a blueshift with increasing  $d$ . We attribute this to two effects: a bandgap modulation with strain and a reduction of the direct Coulomb interaction with increasing interlayer spacing. Regarding the first effect, we observe that the bandgap at  $K$  changes by  $-7$  meV ( $+26$  meV) when  $d$  is compressed (expanded) by  $-0.5$  Å ( $+0.5$  Å). The exciton calculations of  $A_{IL}$  show a much larger shift of  $-20$  meV and  $+38$  meV for compression and expansion, respectively. We attribute the missing energetic offsets to the direct Coulomb interaction being increased for decreasing interlayer distance. Both effects are of similar magnitude: 33 meV for the bandgap modulation and 25 meV for the change in Coulomb interaction between the two extrema.

In figure 4(c) we plot the strength of the in-plane optical dipole  $\langle \hat{x}_{xy} \rangle$  for  $D_{IL}$  and  $A_{IL}$  as a function of interlayer spacing. We observe for  $A_{IL}$  an increase by a factor of four with decreasing interlayer distance, which can be explained by an increased overlap between the hole- and electron wavefunctions for this interlayer exciton. In contrast,  $D_{IL}$  becomes counterintuitively brighter when the layers are pulled apart. We observe that for  $d - d_0 = +0.5$  Å, the electron of  $D_{IL}$  fully localizes in the lower MoSSe layer, and as such  $D_{IL}$  becomes of



**Figure 5.** (a) Trion and exciton absorption spectra of bilayer MoSSe–WSe with energy given relative to  $A_{IL}$ . Negatively charged trions in upper, blue panel, excitons in middle, gray panel and positively charged trions in lower, red panel. The configurational weight of the states of interest are displayed in (b)–(g) in the same fashion as in figures 3(b)–(f).

intralayer character, explaining the large increase in brightness.

Finally, we study the effect of adding additional carriers to the system, leading to the formation of charged excitons, i.e. trions. In this work, we focus only on trion species, where the total center-of-mass momentum of the constituent carriers,  $Q_{COM}$ , equals  $K$  for negatively charged trions and  $K'$  for positively charged trions. This typically results in trion species where all carriers are located in the  $K$ - and/or  $K'$ -valleys. However, for electron doping, lower lying bright and dark states may be found for  $Q_{COM} = Q$  [26, 32], due to the energetically low-lying nature of the  $Q$ -valley in bilayer TMD-like systems. A similar statement holds true for hole doping and the  $\Gamma$ -point. Nevertheless, we do not compute these states, since they are expected to show similar traits regarding bright–dark ordering as for  $Q_{COM} = K$ .

Figure 5(a) shows the absorption spectra of excitons (middle panel), negative trions (upper panel) and positive trions (lower panel). The initial state for photon absorption is represented by an electron located in the lowest conduction band at  $K$  for negatively charged trions, and a hole located in the top valence band at  $K'$  for positively charged trions.  $A_0^-$  is the energetically lowest negatively charged trion state with a binding energy of 11 meV with respect to  $A_{IL}$ , and resembles an electron at  $K$  combined with an  $A_{IL}$ -like electron–hole pair at  $K'$ , as seen in figure 5(b).

We observe three nearly degenerate states, denoted as  $D_{0,1,2}^-$ , positioned 32 meV below  $D_{IL}$  (be reminded that this value corresponds to the fully converged extrapolated value, while the results in figure 5 are for the  $45 \times 45$   $k$ -point grid). This value aligns closely with the trion binding energies of intralayer states identified in Janus TMD monolayers [33], as well as those reported in conventional TMD heterobilayers [14, 34]. We attribute the near-degeneracy of the three states to the spatial separation of wavefunctions, which, due to reduced electron–hole overlap, results in significantly reduced exchange-couplings. In the monolayer, these three states have significantly larger exchange-couplings and, hence, splitting [26]. Moreover,  $D_1^-$  and  $D_2^-$  are deceptively bright—in a similar manner to  $D_{IL}$ . For  $D_1^-$ , we attribute this to the electron localizing slightly in the lower MoS<sub>2</sub> layer, such that a significant overlap forms between the wavefunctions of the two carrier types. In the case of  $D_2^-$ , the relative brightness of the state can be understood from the bright  $A_{IL}$ -like transition, which is spin-allowed but spatially indirect, between the upper valence band and lower conduction band. As opposed to  $D_1^-$  and  $D_2^-$ ,  $D_0^-$  appears completely dark, since we have assumed a thermalization of the additional electron before absorption leaving the upper conduction band empty. At higher temperatures, however, the second conduction band is thermally accessible, in which case  $D_0^-$  is expected to show some bright character. Lastly, the splitting between  $A_0^-$  and  $D_{0,1,2}^-$  is found to be 43 meV—almost identical to the conduction band splitting of 40 meV, and can be easily understood from the configurational weights, see figures 5(c), (e) and (f).

$A_0^+$ , depicted in figure 5(d), is the positively charged analogue of  $A_0^-$  with a binding energy of 11 meV, and also features similar brightness. Contrastingly,  $D_0^+$ , which has the electron in the second conduction band, appears dark, especially compared to the similar case of  $D_{IL}$  and  $D_1^-$ . However, as opposed to the aforementioned states,  $D_0^+$  is a true interlayer state, and is as such almost completely dark.

### 3. Conclusions

Based on *ab initio* screened configuration interaction calculations, we have demonstrated the presence of spin-allowed exciton and trion ground states in MoS<sub>2</sub>–WSe<sub>2</sub> and WSe<sub>2</sub>–WSe<sub>2</sub> bilayers of Janus TMDs. The intra- an interlayer polarization of the Janus bilayers allows, for the specific material combinations mentioned, to shift the bands in such a way that the exciton (and trion) ground state is spin-allowed, in contrast to bilayers of conventional TMDs, where a dark spin-forbidden ground state prevails.

The resulting strong coupling of the ground state to the light field paves the way for faster optoelectronic devices and facilitates the generation, manipulation, and detection of complex excitonic states—including trions—which are especially valuable for valleytronics, spintronics, and advanced quantum information protocols.

Furthermore, we show that both intrinsic structural asymmetry and extrinsic factors such as applied strain can be used as design parameters, enabling the rational creation of heterostructures with tailored spin and optical properties. In particular, our results demonstrate that while the intrinsic Janus asymmetry can lead to bright ground state excitons and trions, the application of external strain provides additional control over their brightness, inspiring new strategies for next-generation quantum light sources and highly responsive 2D optoelectronic platforms.

### 4. Methodology

We employ a parameter-free, truly *ab initio* effective many-body Hamiltonian,  $\mathcal{H}$ , formulated in the electron–hole (eh) picture, adapted from earlier work on semiconductor quantum dots [35, 36] and refined for 2D materials [26, 37]. The Hamiltonian takes the form:

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{eh} + \mathcal{H}_{ee} + \mathcal{H}_{hh},$$

where  $\mathcal{H}_0$  incorporates single-particle energies, while  $\mathcal{H}_{eh}$ ,  $\mathcal{H}_{ee}$  and  $\mathcal{H}_{hh}$  capture electron–hole, electron–electron and hole–hole interactions, respectively, formulated in second quantization. Two-body matrix elements are computed from wavefunctions obtained via density functional theory (DFT) [38, 39], where all components except electron–hole exchange elements include a screened Coulomb potential. A detailed theoretical description can be found in previous works [26]. We obtain the screened interaction kernel,  $W$ , through the random phase approximation [40, 41], while wavefunctions and Kohn–Sham energy levels are computed within the generalized-gradient approximation with PBE parametrization [42]. During structural relaxation of our systems, we include an *ab initio* DFT-D3-BJ parameterization [43] to account for the vdW interaction between the layers. The results presented in this work do not feature self-energy corrected quasiparticle energies as typically found from a GW calculation. This is both due to the extensive nature of GW-convergence, and because our interests do not lie in obtaining energies that are accurate on an absolute energy scale, but rather on the energetic positions of excited states relative to the bandgap and one another [44]. Further computational details and convergence studies are available in the SM.

## Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Supplementary material available at <https://doi.org/10.1088/2053-1583/ae04fb/data1>.

## Acknowledgments

The project is supported by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) within the Priority Program SPP2244 2DMP and by the Cluster of Excellence ‘Advanced Imaging of Matter’ of the DFG—EXC 2056 – Project ID 390715994. Calculations were carried out on Hummel funded by the DFG – 498394658.

## Code availability

The underlying code for this study is not publicly available but may be made available to qualified researchers on reasonable request from the corresponding author.

## Conflict of interest

All authors declare no financial or non-financial competing interests.

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